

Terrain-Aware Semantic Segmentation for Autonomous Mars Rover Driving:

CNNs vs. Transformers vs. Foundation Models

Midpoint Research Check-In

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Abstract

Reliable terrain perception is a prerequisite for longer and safer autonomous Mars rover drives. This project studies four-class semantic segmentation of rover navigation-camera imagery into soil, bedrock, sand, and big rock using the public AI4Mars dataset. Although prior work established that learned models can classify Martian terrain, a controlled and significance-tested comparison of convolutional networks, vision transformers, and foundation-model references—including transfer learning and cross-rover generalization—remains useful. The study compares U-Net and DeepLabV3+ convolutional systems, SegFormer transformers, a small learned reference, a DINOv3 model pretrained on Earth satellite imagery, and a Segment Anything mask-quality upper bound. All trained systems use a common data protocol, loss, evaluation set, and reproducibility record. The primary outcome is mean Intersection-over-Union, supplemented by per-class IoU and descriptive accuracy and boundary measures. Hypotheses are evaluated with a preregistered paired seed/image bootstrap and Holm correction, while cross-rover robustness is assessed by training on Curiosity and testing on Opportunity/Spirit imagery. At this midpoint, 17 full-GPU runs provide descriptive evidence: seven configurations have two completed seeds and three have one. Available-seed mean mIoU is approximately 0.80–0.82 for the conventional systems, compared with 0.681 for Tiny U-Net; a single DINOv3-SAT seed reaches 0.840. Big-rock IoU remains substantially lower than the common terrain classes. These values establish feasibility and motivate the remaining work, but confirmatory H0–H5 decisions remain deferred until the complete three-seed experiment set is available.

1 Introduction

Planetary rovers operate under communication delay, intermittent contact, limited bandwidth, and strict safety constraints. A rover that can recognize the surface immediately ahead can select safer paths and execute longer traverses without waiting for detailed ground commands. Pixel-level terrain understanding is therefore an enabling perception capability: broad soil may be benign, sand may create mobility risk, bedrock can produce wheel wear, and large rocks may be direct hazards. The consequences are concrete. Curiosity accumulated wheel damage while traversing sharp embedded rocks, motivating stronger onboard terrain assessment and systems such as Soil Property and Object Classification (SPOC) [3].

AI4Mars makes a systematic machine-learning study of this task possible. Its merged public release contains crowdsourced terrain annotations for rover imagery and smaller expert-agreement

test sets [1, 2]. This project focuses on grayscale Curiosity navigation-camera images because they are the stereo imagery used for driving and the release provides an expert test split for that camera. A separate expert-labeled collection from the Mars Exploration Rovers Opportunity and Spirit permits a stringent test of camera and terrain distribution shift.

The applied problem is dense multi-class semantic segmentation. Unlike image classification, the model must preserve fine boundaries around rocks while also recognizing large, visually uniform terrain regions. This tension motivates a comparison among three families of visual inductive bias. U-Net and DeepLabV3+ use local convolution, skip connections, and multi-scale receptive fields [6, 7]. SegFormer uses hierarchical self-attention to capture broader context [8]. Foundation models bring priors learned from far larger collections, although the difference between Earth imagery and ground-level Mars imagery makes their transfer uncertain [9, 10].

The central research question is:

For Mars terrain segmentation, how do convolutional, transformer, and foundation-model systems compare; how much does pretrained initialization help; and how well does a Curiosity-trained system generalize to Opportunity/Spirit imagery?

The project makes four intended contributions. First, it provides a leakage-safe, reproducible AI4Mars pipeline with explicit ignore-mask handling. Second, it compares representative CNN and transformer systems under a shared training and evaluation protocol. Third, it measures transfer learning and cross-rover generalization rather than reporting only a single in-domain score. Fourth, it binds scientific claims to preregistered statistical decision rules. The last contribution is especially important at the midpoint: successful prototype runs demonstrate feasibility, but they are not treated as confirmatory evidence until all required seeds are complete.

2 Related Work

2.1 Mars terrain perception

SPOC demonstrated deep learning for terrain classification in Mars rover missions and connected visual terrain labels to operational mobility needs [3]. Swan et al. introduced AI4Mars, combining rover images with crowdsourced labels and expert validation to support terrain-aware autonomous driving research [1]. The current merged-0.6 release is distributed publicly through Zenodo [2]. More recent Mars-specific segmentation work, including MarsSeg, explores specialized multi-level feature extraction [4]. These studies establish task relevance, but they do not resolve the controlled combination of architecture family, initialization, cross-rover evaluation, and paired inference addressed here.

2.2 Convolutional semantic segmentation

Fully convolutional networks replaced dense classification heads with spatial prediction and laid the groundwork for end-to-end semantic segmentation [5]. U-Net combines a contracting encoder with an expanding decoder and high-resolution skip connections, allowing semantic context and localized boundary detail to interact [6]. DeepLabV3+ instead emphasizes multi-scale context through atrous spatial pyramid pooling and a decoder that refines object boundaries [7]. Residual encoders provide deep, trainable feature hierarchies and widely used ImageNet initialization [11, 13]. These CNNs supply strong locality and translation priors that may be valuable when AI4Mars training data is limited or labels are noisy.

2.3 Transformers and foundation models

The Vision Transformer showed that attention-based models can learn visual representations from sequences of image patches [12]. SegFormer adapts hierarchical transformer features to semantic segmentation with efficient sequence reduction and a lightweight decoder [8]. Its global context may help with broad soil and sand regions, although its boundary behavior and data requirements must be measured rather than assumed.

Segment Anything (SAM) was trained on more than a billion masks and produces class-agnostic regions from prompts [9]. AI4Mars provides no semantic prompt channel, so SAM is not treated here as deployable zero-shot terrain classification. Instead, a preregistered region-oracle assignment measures an explicit upper bound on what a semantic labeler could obtain from SAM’s mask proposals. DINOv3 is a 2025 Meta AI research preprint on self-supervised visual representation learning at scale [10]; it is not described as peer reviewed. Its SAT-493M variant is nevertheless scientifically interesting because it transfers features learned from Earth orbital imagery to ground-level Mars imagery. That cross-planet and cross-viewpoint gap makes the reference a test of representation transfer, not an assumed advantage.

3 Research Project Problem

3.1 Task definition and scope

Let a grayscale navigation-camera image be $\mathbf{x} \in \mathbb{R}^{H \times W}$. It is replicated across three channels for compatibility with pretrained encoders. A segmentation model $f_\theta : \mathbb{R}^{H \times W \times 3} \rightarrow \mathbb{R}^{H \times W \times C}$ produces logits for $C = 4$ terrain classes. The predicted label map is

$$\hat{y}_{ij} = \arg \max_{c \in \{0,1,2,3\}} f_\theta(\mathbf{x})_{ijc}, \quad (1)$$

where the classes are soil, bedrock, sand, and big rock. Ground-truth value 255 denotes ignored pixels, including unlabeled regions, rover hardware, or regions outside the labeling range. Ignored pixels contribute to neither training loss nor evaluation metrics.

The primary in-rover dataset contains 16,064 labeled Curiosity Navcam training images and 322 expert-labeled test images from the pinned `masked-gold-min3-100agree` subset. The training images are split by image into 12,851 training and 3,213 validation records using seed 1414. Curiosity Mastcam imagery is out of scope because it is a different color science camera without the corresponding gold test protocol. The cross-rover test contains 204 expert-labeled Opportunity/Spirit images and is never used for training.

3.2 Research gap

Individual Mars segmentation systems can achieve useful performance, but a reported score alone does not identify why it is high or whether it will transfer. Architecture, pretrained initialization, class imbalance, seed variation, and rover/camera domain shift may all influence an observed result. The project addresses this gap with controlled within-architecture initialization comparisons, configured cross-family comparisons, per-class analysis, and paired uncertainty estimation over both training seed and test image.

3.3 Hypotheses

The executable hypothesis definitions are frozen in `configs/hypotheses.yaml`; the summary below follows the current protocol rather than superseded proposal wording.

- H0** No tested deep system significantly exceeds the constant-majority reference on mIoU. H0 is rejected exactly when H1 is supported; otherwise the study reports failure to reject, not evidence that the systems are equivalent.
- H1** At least one conventional deep system—U-Net, DeepLabV3+, or SegFormer—exceeds the constant-majority reference on mIoU under the Family-A adjusted test.
- H2** Pretrained initialization improves at least one tested architecture relative to its backbone-matched from-scratch counterpart. The inference is deliberately bounded and does not claim that pretraining helps every architecture.
- H3** Configured SegFormer MiT-B2 systems differ from the configured U-Net with ResNet-34 or DeepLabV3+ with ResNet-50 systems overall or for at least one terrain class. Effects are reported as configured-system differences, not as a pure causal effect of attention.
- H4** The validation-selected conventional system has a Curiosity-to-MER mIoU drop below 0.15 and a cross-rover 90% confidence-interval lower bound above the MER majority reference.
- H5** DINOv3-SAT or the SAM region-oracle upper bound exceeds the learned Tiny U-Net reference under the gated Family-E comparison. Missing gated weights or incomplete learned seed sets defer this hypothesis.

4 Method

4.1 Data pipeline and leakage controls

The dataset index pairs image and label paths within a single rover/camera subtree and assigns each image a stable, camera-qualified name. Splits operate on those image records rather than on crops, so a frame cannot occur in both training and validation sets. The official expert test images are held out from model and hyperparameter selection. The MER set is test-only. Preflight checks assert the expected record counts, image/label pairing, split disjointness, permitted mask values, and archive/index fingerprints.

Images are decoded as grayscale and replicated to three channels. Training and evaluation resize to 512×512 pixels and use ImageNet normalization. The frozen augmentation recipe allows a horizontal flip and mild brightness/contrast perturbation. It excludes vertical flips because inverting the horizon is physically implausible. Nearest-neighbor mask interpolation preserves categorical labels. Class weights are computed from training-split pixel frequencies and clipped to avoid unstable extremes.

4.2 Compared systems

Table 1: Planned systems and their roles in the preregistered study.

System	Backbone/variant	Initialization	Protocol role
Majority	Constant soil	None	H1/H4 non-learned reference
Tiny U-Net	Two-level CNN	Random	H5 learned reference
U-Net	ResNet-34	ImageNet / random	CNN and H2 comparison
DeepLabV3+	ResNet-50	ImageNet / random	Multi-scale CNN and H2
SegFormer	MiT-B0/B2	ImageNet / random	Transformer and H2/H3
DINOv3-SAT	ViT-L/16 SAT-493M	Self-supervised	Gated H5 reference
SAM	ViT-B proposals	Foundation weights	Region-oracle upper bound

The Tiny U-Net contains two encoder/decoder levels and is intentionally small. U-Net fuses

encoder and decoder features through skip connections. For encoder feature $\mathbf{e}_\ell$ and deeper decoder feature $\mathbf{d}_{\ell+1}$, a decoder stage has the schematic form

$$\mathbf{d}_\ell = \text{Conv}([\text{Up}(\mathbf{d}_{\ell+1}); \mathbf{e}_\ell]). \quad (2)$$

DeepLabV3+ uses atrous convolution to enlarge the receptive field without reducing feature-map resolution. For dilation r ,

$$y[i] = \sum_k x[i + rk]w[k]. \quad (3)$$

SegFormer uses hierarchical Mix-Transformer features. Its attention operation is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V, \quad (4)$$

with sequence reduction to control cost at image resolution.

The DINOv3-SAT arm uses a satellite-pretrained visual backbone with a trained segmentation head. The SAM arm generates class-agnostic proposals. For its oracle upper bound, each proposal receives the majority ground-truth class among its valid pixels; later proposals overwrite earlier ones and uncovered pixels default to soil. Because this procedure consults the target label, it is explicitly an analysis upper bound and cannot be represented as deployable inference.

4.3 Optimization objective

All trained conventional systems minimize class-weighted cross-entropy plus soft Dice loss over valid pixels:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}^w(\hat{\mathbf{y}}, \mathbf{y}) + \lambda(1 - \text{Dice}(\hat{\mathbf{y}}, \mathbf{y})), \quad (5)$$

$$\text{Dice} = \frac{2 \sum p g + \epsilon}{\sum p + \sum g + \epsilon}. \quad (6)$$

The validity mask is applied to both terms. Optimization uses AdamW [15]; validation mIoU controls model selection, and the expert test set is not used to choose an epoch or configuration.

4.4 Evaluation measures

For class c , Intersection-over-Union is

$$\text{IoU}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c}. \quad (7)$$

The primary metric, mIoU, is the macro mean over the fixed set of classes present in the full evaluation split. Per-class IoU is essential because the common soil class can obscure failures on the drivability-relevant big-rock class. Pixel accuracy and boundary-F1 at three-pixel tolerance are reported descriptively but are not hypothesis-tested. Per-image integer intersections, unions, correct pixels, and valid-pixel counts are stored so aggregate metrics can be reconstructed under resampling without retaining predicted masks.

4.5 Statistical protocol

Learned confirmatory arms require training seeds 1414, 1415, and 1416. Deterministic references are computed once and paired against each learned seed. The primary inference uses 10,000 paired bootstrap resamples, first over training seeds and then over images within a seed, with purpose seed 0 reset per comparison. The study reports percentile 90% confidence intervals and plus-one p-values. Family A and B tests are one-sided in their preregistered positive directions; Family C is two-sided and includes overall and fixed-set per-class comparisons in one correction family. Holm’s procedure controls multiplicity within each hypothesis family at $\alpha = 0.10$ [17]. H4 uses its preregistered threshold-plus-interval rule rather than a p-value.

Every run records its resolved configuration, seed, git revision, configuration/code hash, package versions, hardware profile, parameter counts, timestamps, and per-image sufficient statistics. This manifest-based design separates reproducibility evidence from large untracked checkpoints and raw data.

4.6 End-to-end pipeline

```
AI4Mars MSL Navcam images + labels (255 = ignore)
|
camera-scoped index -> by-image train/validation split
|
resize, safe augmentation, normalization, class weights
|
+-----+-----+-----+-----+
| U-Net   | DeepLabV3+ | SegFormer | DINOv3 / SAM |
+-----+-----+-----+-----+
|
weighted cross-entropy + Dice (valid pixels only)
|
MSL expert test + MER cross-rover test
|
per-image counts -> paired seed/image bootstrap -> Holm
```

5 Experimental Progress and Collection Plan

The implementation and bounded notebook prototype establish that images, labels, transforms, models, loss, and metrics can be connected. The current result store contains 17 successful full-GPU in-rover mIoU records spanning ten configurations. Seven configurations have completed seeds 1414 and 1415; SegFormer B2 pretrained, SegFormer B2 scratch, and DINOv3-SAT currently have seed 1414 only. Table 2 makes the remaining work explicit. An *complete* entry means that a test artifact and per-image sufficient statistics exist, not that the comparison is confirmatory.

6 Statistical Analysis and Results

6.1 Preliminary descriptive results

Table 3 reports the evidence currently available. For two-seed configurations, mIoU is the arithmetic mean with the sample standard deviation across seeds in parentheses. Per-class values are available-

Table 2: Midpoint full-GPU run inventory. Both initialization variants have the same status where shown as a pair. Seed 1416 is required before confirmatory inference.

System	Variant	1414	1415	1416
Tiny U-Net	scratch	complete	complete	pending
U-Net R34	pretrained / scratch	complete	complete	pending
DeepLabV3+ R50	pretrained / scratch	complete	complete	pending
SegFormer B0	pretrained / scratch	complete	complete	pending
SegFormer B2	pretrained / scratch	complete	pending	pending
DINOv3-SAT	finetuned	complete	pending	pending
SAM oracle	deterministic	pending	—	—

seed means. DINOv3-SAT has only one completed seed, so neither a seed standard deviation nor a fair stability comparison is yet possible. These are descriptive summaries; the preregistered hierarchical bootstrap is not approximated with a two-seed test.

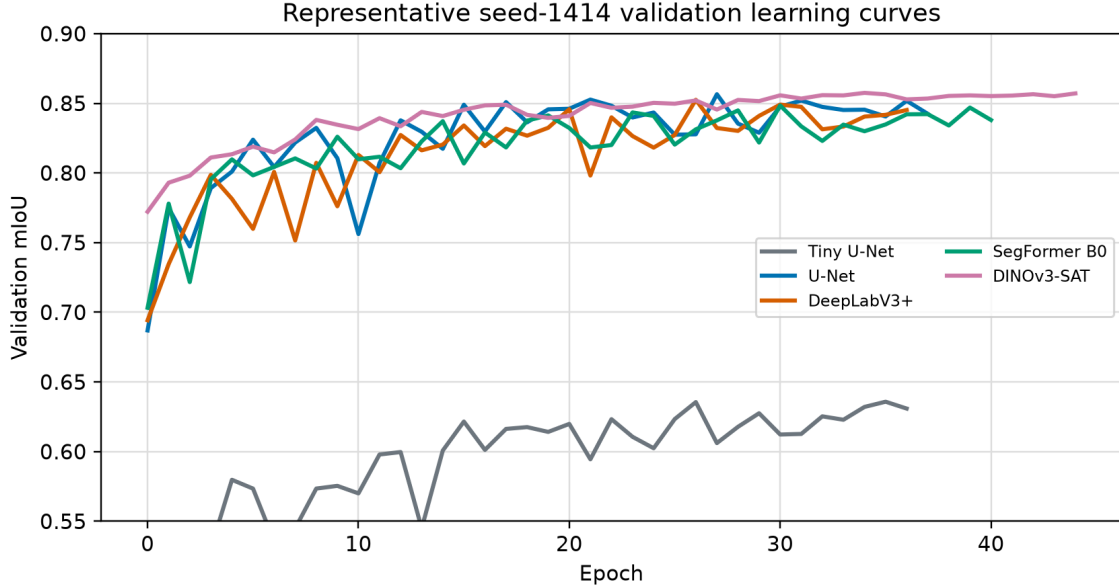
Table 3: Preliminary in-rover expert-test performance. SD is across available training seeds; the DINOv3-SAT entry is a single run.

System	n	mIoU mean (SD)	Soil	Bedrock	Sand	Big rock
Tiny U-Net scratch	2	0.681 (0.002)	0.918	0.837	0.873	0.094
U-Net R34 pretrained	2	0.821 (0.018)	0.979	0.942	0.962	0.399
U-Net R34 scratch	2	0.822 (0.001)	0.983	0.933	0.962	0.409
DeepLabV3+ R50 pretrained	2	0.817 (0.022)	0.982	0.937	0.966	0.382
DeepLabV3+ R50 scratch	2	0.819 (0.012)	0.980	0.940	0.967	0.388
SegFormer B0 pretrained	2	0.816 (0.016)	0.981	0.940	0.964	0.379
SegFormer B0 scratch	2	0.802 (0.004)	0.960	0.897	0.908	0.444
DINOv3-SAT finetuned	1	0.840 (—)	0.981	0.940	0.970	0.469

Across the two-seed conventional configurations, mean mIoU lies in a narrow band from 0.802 to 0.822. U-Net scratch has the largest available-seed mean (0.822), but it differs from pretrained U-Net by only 0.001 and seed 1414 and 1415 favor different variants. DeepLabV3+ shows the same directionally inconsistent pattern. SegFormer B0 is the only two-seed architecture for which the available mean favors pretraining (0.816 versus 0.802). These mixed directions are exactly why H2 must remain unresolved until seed 1416 and the paired test are complete.

The single DINOv3-SAT run has the highest observed point estimate, 0.840, and the highest currently observed big-rock IoU among the representative systems, 0.469. This is promising evidence for continuing H5, not evidence that H5 is supported. Across every representative system, soil, bedrock, and sand IoUs are high, while big-rock IoU is substantially lower. The class-level pattern suggests that rare hazard segmentation, rather than broad terrain labeling, is the principal remaining error mode. Figure 2 illustrates both strong broad-region agreement and visibly different handling of the large-rock boundary.

Midpoint placeholder—not a result. Populate the cross-rover table with the validation-selected conventional system, its MSL and MER mIoU, their difference, the MER hierarchical confidence interval, and the MER majority score. Apply the exact H4 threshold rule without substituting a post hoc test.



Curves document pipeline convergence; final comparisons use canonical three-seed test artifacts.

Figure 1: Representative seed-1414 validation learning curves. These curves demonstrate stable optimization and model-selection feasibility; they are not hypothesis tests.

6.2 Confirmatory analysis still pending

Table 4: Preregistered hypothesis tests (confirmatory placeholder).

Hypothesis	Comparison	Effect	90% CI	Holm p	Decision
H1	deep vs. majority	TBD	TBD	TBD	deferred
H2	pretrained vs. scratch	TBD	TBD	TBD	deferred
H3	SegFormer vs. CNNs	TBD	TBD	TBD	deferred
H4	MSL-to-MER drop	TBD	TBD	n/a	deferred
H5	foundation vs. Tiny U-Net	TBD	TBD	TBD	deferred

At the midpoint, every hypothesis remains deferred. The descriptive results above document progress without converting partial seed sets into adjusted p-values or hypothesis decisions. Final inference will be generated only after canonical selection verifies the required run set and the executable protocol snapshot matches the preregistration.

7 Discussion and Limitations

Midpoint placeholder—not a result. Interpret only supported, multiplicity-corrected effects. Discuss whether observed differences are practically meaningful for terrain classes, not merely statistically detectable. Relate overall results to parameter counts and per-class/boundary behavior without turning the configured-system comparisons into architecture-only causal claims.

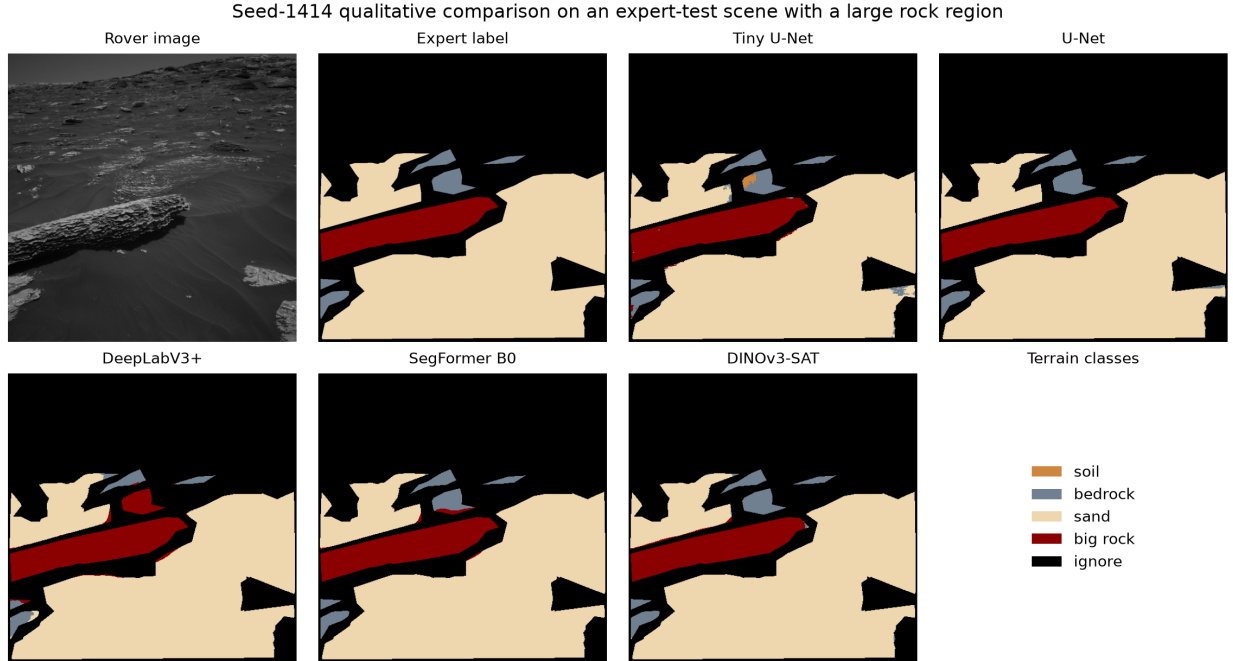
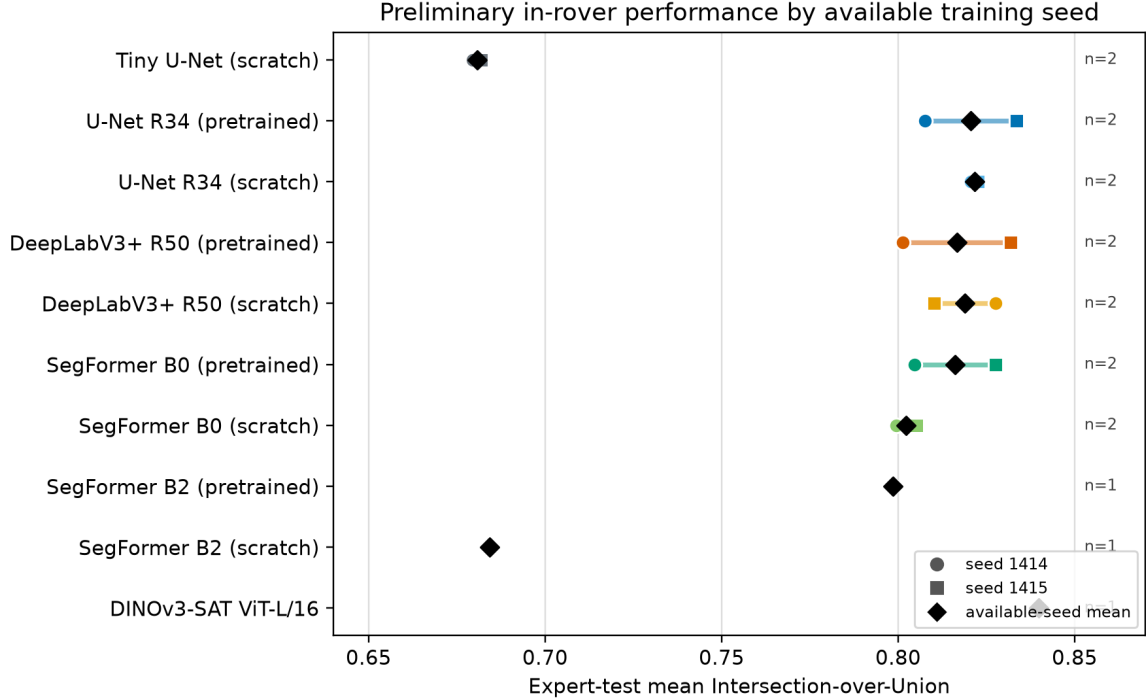


Figure 2: A fixed expert-test scene containing a large rock region, shown with the expert label and seed-1414 predictions. Black pixels are ignored. The example was selected for rare-class coverage, not because it maximized any model’s score.

The preliminary evidence already identifies useful questions for the final discussion. First, conventional architecture point estimates are much closer to one another than they are to Tiny U-Net, so uncertainty and paired effects will matter more than rank ordering. Second, the available pre-training effects are small and inconsistent across seeds for the CNNs. Third, the highest single-seed point belongs to DINOv3-SAT, but its stability is unknown. Finally, all systems show a large gap between common terrain and big rock. This gap has direct operational relevance because a rare large rock can matter more to a rover path than another correctly labeled patch of soil.

Several limitations are known before results. AI4Mars labels combine crowdsourced judgments and expert-agreement subsets, so both class ambiguity and annotation noise matter. The primary scope is grayscale Navcam imagery and does not establish performance on Mastcam, orbital imagery, or future rovers. MER tests domain shift but cannot identify whether any degradation arises from camera, mission, lighting, geography, or all four. Big rock is rare, making its estimate more variable than common-class estimates. Three training seeds improve uncertainty accounting but do not exhaust the space of optimization outcomes. Finally, SAM’s region-oracle assignment uses target labels and is only an upper bound; it cannot be presented as an autonomous semantic-segmentation system.



Descriptive midpoint results only; seed 1416 and preregistered paired inference are pending.

Figure 3: Every available full-GPU in-rover mIoU result. Points show individual seeds; black diamonds show the mean of the currently available seeds. Connecting lines are descriptive ranges, not confidence intervals.

8 Conclusion

Midpoint placeholder—not a result. State the final answer to the research question after H0–H5 are evaluated. Summarize which configured system performs best, whether transfer helps any architecture, which classes show material differences, and whether cross-rover degradation meets the preregistered bound. Connect the evidence to rover drivability and identify the next most defensible experiment.

This midpoint establishes the research problem, literature position, controlled method, runnable prototype, descriptive results from 17 full-GPU runs, and statistical collection plan. The current evidence shows that the pipeline learns strong broad-terrain segmentations and that rare big-rock recognition remains difficult. It intentionally does not convert incomplete experiment runs into confirmatory scientific conclusions.

Data and Code Availability

The AI4Mars merged-0.6 archive is publicly available from Zenodo record 15995036 [2]. The expected archive is approximately 16.2 GB with MD5 `daf80a86021253292e6c425f97baa5c6`. It is not included with this submission. The accompanying notebook provides download, integrity-check, and expected-layout instructions and uses the project code directly. Raw data, model weights, secrets, checkpoints, and generated prediction masks remain excluded from version control.

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